NIRBHIK GUPTA

BS-MS Dual Degree in Physics, IISER Mohali

Interests: quantum technologies, computational physics, nonlinear optics and fibre photonics

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Education

Indian Institute of Science Education and Research (IISER) Mohali Aug 2024 to Present
BS-MS Dual Degree in Physics *Mohali, India*

- POSE Scholarship (Promotion of Science Education), Government of Haryana, 2024

Research Experience

Research Intern May 2026 to Present
Indian Institute of Technology (IIT) Bombay, under Prof. Arun Surendran *Hybrid*

- Studying parametric four-wave mixing (FWM) and supercontinuum generation in optical fibres, focusing on dispersion, energy conservation, and phase-matching conditions.
- Cut the runtime of a Python FWM simulation from about 2 days to 1 hour through vectorisation, a better algorithm, and a physics derivation worked out step by step. The code scans roughly 300 million frequency pairs and returns about 200,000 valid combinations.
- Learning fibre cleaving and splicing, and using an Optical Spectrum Analyser (OSA) to characterise the supercontinuum spectra the fibres generate.

Research Intern May 2025 to Jul 2025
IISER Mohali, under Prof. Sudeshna Sinha *Mohali, India*

- Investigated deterministic chaos and nonlinear dynamics: discrete and continuous systems, bifurcation theory, and sensitivity to initial conditions.
- Studied Logistic, Tent, and Shift maps, Bernoulli shifts, linear stability, and the effect of finite-precision arithmetic on chaotic dynamics.
- Developed Python simulations of the Lorenz and Rössler attractors using numerical integration, and analysed intermittency and transitions in nonlinear systems.

Skills

- **Programming:** Python, C++, MATLAB, LaTeX, Linux (Bash), Git/GitHub
- **Scientific computing:** NumPy, SciPy, Matplotlib, Pandas, Scikit-Learn, TensorFlow
- **Numerical methods:** ODE solvers (Runge-Kutta, Euler), time-series analysis, linear stability analysis
- **Optics and photonics:** Fibre cleaving and splicing, supercontinuum generation, spectroscopy, OSA
- **Languages:** English, Hindi, Punjabi, French

Relevant Coursework

Physics: Mechanics, Electromagnetism, Waves and Optics, Thermodynamics and Statistical Physics, Modern Physics, Mathematical Methods in Physics, Quantum Mechanics
Mathematics: Linear Algebra, Probability and Statistics, Calculus, Number Theory, Sequences and Series

Schools and Workshops

IQM Quantum Summer School 2026 25 to 27 Aug 2026
Practical sessions on quantum algorithms and superconducting hardware. Built and ran circuits (Bernstein-Vazirani, Grover's search, QAOA) on IQM Resonance, and studied pulse-level control and noise.

QFTA 2025: Quantum Foundations and Technological Applications IISER Mohali
Workshops on quantum measurement, entanglement, metrology and sensing, and QKD.

Leadership and Activities

- Polymathia, Tattva Science Fest (2025, 2026): helped organise and run the event at IISER Mohali.
- Insomnia, Cultural Fest (2026): sponsorship outreach and financial coordination.
- IISER Mohali 20th Foundation Day (2026): member of the organising team.
- Incoming Batch Registration (2026): Convener of the team managing registration for the new batch.

References

Prof. Sudeshna Sinha
Professor, Department of Physics, IISER Mohali
sudeshna@iisermohali.ac.in

Prof. Arun Surendran
Assistant Professor, PhInLAB, IIT Bombay
sarun@iitb.ac.in